

Murat Keçeli

February 2026

Argonne National Laboratory
Computational Science Division
9700 S. Cass Avenue Lemont, IL 60439

<http://keceli.github.io>
keceli@anl.gov
+1 630 252 5695

Professional Experience

- **Argonne National Laboratory** Lemont, IL
Computational Scientist Present
- **Argonne National Laboratory** Lemont, IL
Postdoctoral Appointee 2014 - 2018
 - Supervisors: Dr. Albert F. Wagner, Dr. Stephen J. Klippenstein
- **Massachusetts Institute of Technology** Cambridge, MA
Postdoctoral Associate 2012 - 2014
 - Supervisor: Prof. William H. Green

Education

- **University of Illinois at Urbana-Champaign** Urbana, IL
Ph.D in Chemical Physics May 2012
 - Supervisor: Prof. So Hirata
 - Thesis title: Vibrational Many-Body Methods for Molecules and Extended Systems
- **Bilkent University** Ankara, Turkey
B.Sc. & M.Sc. in Physics 2004 & 2006
 - Supervisor: Prof. Mehmet Ö. Oktel
 - Thesis title: A Variational Approach to Stationary and Rotating Bose-Einstein Condensates

Honors and Awards

- Recipient of Impact Argonne awards in 2023 and 2025.
- 2022 Physical Sciences and Engineering (PSE) Excellence Award for Scientific Achievement by Argonne National Laboratory, recognized for the collaborative effort in the development of the Auto-Mech software suite.
- Distinguished paper award for “Automated theoretical chemical kinetics: Predicting the kinetics for the initial stages of pyrolysis” by the 38th International Symposium on Combustion in 2021.
- Graduate student fellowship by the University of Illinois at Urbana-Champaign in 2011-2012.
- Conference travel awards by the University of Florida in 2010 and by the University of Illinois at Urbana-Champaign in 2011 and 2012.
- First place in the poster presentation competition at the 18th conference on Current Trends in Computational Chemistry in Jackson, Mississippi in 2009.
- Merit based comprehensive scholarship (covers tuition, accommodation, and monthly salary) during B.Sc. and M.Sc. studies at Bilkent University.

Current and Past Funding

- Benchmarking and Design of Agentic Foundation Models for the Accelerated Materials Discovery of Selective Ion Absorbers for Water and Recovery Applications
 - Role: Co-PI
 - Agency: LDRD
 - Status: Ended
 - Period: 06/2025 - 10/2025
 - Funding amount: \$20,000 of total \$100,000
- FireAID: An Undergraduate Research Training Program to Develop Technologies to Fight Wildland Fire with Artificial Intelligence and Deep-Learning in Alaska
 - Role: ANL Lead PI
 - Agency: DOE/ASCR
 - Status: Ongoing
 - Period: 09/2025 - 09/2028
 - Funding amount: \$450,000 of total \$1,250,000
- Accelerated Innovation of Advanced Membranes for CO₂ Capture and Conversion Guided by Artificial Intelligence via Inverse Design
 - Role: Co-PI
 - Agency: LDRD
 - Status: Ongoing
 - Period: 10/2024 - 09/2026
 - Funding amount: \$50,000 of total \$250,000
- Photocarboxylation: From Captured CO₂ to Value-added Chemicals with Metal Organic Frameworks
 - Role: Co-PI
 - Agency: DOE/BES
 - Status: Ongoing
 - Period: 10/2024 - 09/2026
 - Funding amount: \$200,000 of \$2,200,000
- Center for Steel Electrification by Electrosynthesis (C-STEEL)
 - Role: Collaborator
 - Agency: DOE
 - Status: Ended
 - Period: 06/2024 - 09/2024
 - Funding: 20% FTE
- AuroraGPT: Foundation Models for Science

- Role: Collaborator
- Agency: LDRD
- Status: Ended
- Period: 09/2024 - 09/2026
- Funding: \$40,000
- NWChemEx:
 - Role: ANL Lead PI (3/2022 - 12/2023), Key personnel (10/2019 - 3/2022).
 - Agency: ECP/ASCR
 - Status: Ended
 - Period: 10/2019 - 12/2023
 - Funding amount: \$2 million (total)
- Autonomous Discovery for Science, Automated Framework for Carbon Deconstruction:
 - Role: Co-PI
 - Agency: LDRD/PSE
 - Status: Ended
 - Period: 06/2022 - 09/2024
 - Funding amount: \$60,000 (FY24), \$50,000 (FY23), \$50,000 (FY22)
- Neural Network Potentials for Simple Alkanes:
 - Role: PI
 - Agency: LDRD/ALCF
 - Status: Ended
 - Period: 05/2020 - 09/2020
 - Funding amount: \$21,500

Professional Activities

- ALCF Allocations committee member, 2024-2025.
- ALCF Workshops organization committee member, 2023-2025.
- Co-organizer of AI for Chemistry and Materials Hands-on Series, 2023.
- ALCF software committee member, 2020-2023.
- ATPESC review committee member, 2018-2025.
- Guest associate editor of Frontiers in Theoretical and Computational Chemistry. Created a special issue on Challenges and Opportunities for Computational Chemistry in the Exascale Era, 2022-2024.
- Session chair for MolSSI Workshop on HPC in Computational Chemistry and Materials Science, 2021.
- Member of American Physical Society since 2004, American Chemical Society since 2010, and Association for Computing Machinery since 2023.

Journal Articles (h-index=19, citations>1000)

1. T. D. Pham, A. Tanikanti, and M. Keçeli, "ChemGraph as an agentic framework for computational chemistry workflows," *Communications Chemistry*, 2026
2. T. Kocabas, M. Keceli, T. Gurel, M. Milosevic, and C. Sevik, "Thermal conductivity limits of MoS₂ and MoSe₂: Revisiting high-order anharmonic lattice dynamics with machine learning potentials," *arXiv preprint arXiv:2509.13798*, 2025
3. R. Reji, J. P. Vizuet, X. Yan, B. K. Behera, X. Zheng, M. Klenk, N. P. Weingartz, G. T. Kostopoulos, L. X. Chen, P. Zapol, *et al.*, "Photochemical co₂ reduction by a postsynthetically modified zr-mof," *Inorganic Chemistry*, 2025
4. J. Wen, S. Nalawade, Z. Liang, C. Bielick, M. F. Boston, A. Chowdhury, A. Collin, L. D. Angelis, J. Ellen, H. Frase, R. R. Gameiro, J. M. Gutierrez, P. Kadam, M. Keceli, S. Krishnamurthy, A. Kwok, Y. L. Lu, H. Mattie, L. G. McCoy, K. Miller, A. C. Morgan, M. L. Moerig, T. Nguyen, A. Owen-Post, A. D. Ruiz, S. R. Puchala, S. Samineni, T. Tohyama, V. Ullanat, C. Valenza, C. Velez, P. Wang, A. Wuest, Y. Zhou, Y. Zhu, J. M. Johnson, N. Lenane, J. Willcox, F. J. Vitiello, L. A. G. Celi, and R. Umeton, "Red teaming for generative ai, report on a copyright-focused exercise completed in an academic medical center," *arXiv preprint arXiv:2506.22523*, 2025
5. B. Smith, J. Zhang, H. Zhang, L. C. McInnes, M. Keceli, A. Vasan, S. Balay, T. Isaac, L. Chen, and V. Vishwanath, "Ai assistants to enhance and exploit the petsc knowledge base," *arXiv preprint arXiv:2506.20608*, 2025
6. R. Reji, J. P. Vizuet, X. Yan, B. Behera, X. Zheng, M. Klenk, N. P. Weingartz, G. T. Kostopoulos, L. X. Chen, N. Agarwal, P. Zapol, E. A. Huerta, M. Keçeli, and K. Glusac, "Product selectivity in photochemical co₂ reduction by a post-synthetically modified zr-mof," *ChemRxiv*, 2025
7. F. Cappello, S. Madireddy, R. Underwood, N. Getty, N. L.-P. Chia, N. Ramachandra, J. Nguyen, M. Keçeli, T. Mallick, Z. Li, *et al.*, "Eaira: Establishing a methodology for evaluating ai models as scientific research assistants," *arXiv preprint arXiv:2502.20309*, 2025
8. M. A. Leung, K. Cahill, R. Hartman-Baker, P. Kinsley, L. C. McInnes, S. Parete-Koon, S. Abraham, L. B. Barrier, G. Chen, L. DeStefano, M. Keçeli, *et al.*, "Communities through energy justice projects," *Journal of Computational Science*, vol. 15, no. 1, 2024
9. R. M. Richard, K. Keipert, J. Waldrop, M. Keçeli, D. Williams-Young, R. Bair, J. Boschen, Z. Crandall, K. Gasperich, Q. I. Mahmud, *et al.*, "Pluginplay: Enabling exascale scientific software one module at a time," *The Journal of Chemical Physics*, vol. 158, no. 18, 2023
10. T. Kocabas, M. Keçeli, Á. Vázquez-Mayagoitia, and C. Sevik, "Gaussian approximation potentials for accurate thermal properties of two-dimensional materials," *Nanoscale*, vol. 15, no. 19, pp. 8772–8780, 2023
11. S. N. Elliott, M. Keçeli, M. K. Ghosh, K. P. Somers, H. J. Curran, and S. J. Klippenstein, "High-accuracy heats of formation for alkane oxidation: From small to large via the automated cbh-anl method," *The Journal of Physical Chemistry A*, vol. 127, no. 6, pp. 1512–1531, 2023
12. S. N. Elliott, K. B. Moore III, A. V. Copan, Y. Georgievskii, M. Keçeli, K. P. Somers, M. K. Ghosh, H. J. Curran, and S. J. Klippenstein, "Systematically derived thermodynamic properties for alkane oxidation," *Combustion and Flame*, p. 112487, 2023
13. G. D. Fletcher, C. Bertoni, M. Keçeli, and M. J. DMello, "Prediction of correlation energies using variational subspace valence bond," *The Journal of Chemical Physics*, vol. 157, no. 12, 2022

14. I. Demiroğlu, Y. Karaaslan, T. Kocabaş, M. Keçeli, A. Vazquez-Mayagoitia, and C. Sevik, "Computation of the thermal expansion coefficient of graphene with gaussian approximation potentials," *The Journal of Physical Chemistry C*, vol. 125, no. 26, pp. 14409–14415, 2021
15. J. Ruiz, K. Misa, A. Seshappan, M. Keçeli, and O. Sode, "Exploring the anharmonic vibrational structure of carbon dioxide trimers," *The Journal of Chemical Physics*, vol. 154, no. 14, p. 144302, 2021
16. K. Kowalski, R. Bair, N. P. Bauman, J. S. Boschen, E. J. Bylaska, J. Daily, W. A. de Jong, T. Dunning Jr, N. Govind, R. J. Harrison, *et al.*, "From NWChem to NWChemEx: Evolving with the computational chemistry landscape," *Chemical reviews*, vol. 121, no. 8, pp. 4962–4998, 2021
17. S. N. Elliott, K. B. Moore III, A. V. Copan, M. Keçeli, C. Cavallotti, Y. Georgievskii, H. F. Schaefer III, and S. J. Klippenstein, "Automated theoretical chemical kinetics: Predicting the kinetics for the initial stages of pyrolysis," *Proceedings of the Combustion Institute*, vol. 38, no. 1, pp. 375–384, 2021
18. R. Vescovi, H. Li, J. Kinnison, M. Keçeli, M. Salim, N. Kasthuri, T. D. Uram, and N. Ferrier, "Toward an automated hpc pipeline for processing large scale electron microscopy data," in *2020 IEEE/ACM 2nd Annual Workshop on Extreme-scale Experiment-in-the-Loop Computing (XLOOP)*, pp. 16–22, IEEE, 2020
19. V. W.-z. Yu, C. Campos, W. Dawson, A. García, V. Havu, B. Hourahine, W. P. Huhn, M. Jacquelin, W. Jia, M. Keçeli, *et al.*, "ELSI - an open infrastructure for electronic structure solvers," *Computer Physics Communications*, vol. 256, p. 107459, 2020
20. W. Dong, M. Keçeli, R. Vescovi, H. Li, C. Adams, E. Jennings, S. Flender, T. Uram, V. Vishwanath, N. Ferrier, *et al.*, "Scaling distributed training of flood-filling networks on hpc infrastructure for brain mapping," in *2019 IEEE/ACM Third Workshop on Deep Learning on Supercomputers (DLS)*, pp. 52–61, IEEE, 2019
21. S. Maystrovsky, M. Keçeli, and O. Sode, "Understanding the anharmonic vibrational structure of the carbon dioxide dimer," *The Journal of Chemical Physics*, vol. 150, no. 14, p. 144302, 2019
22. G. D. Fletcher, C. Bertoni, M. Keçeli, and M. D'Mello, "Valence: A massively parallel implementation of the variational subspace valence bond method," *Journal of Computational Chemistry*, vol. 40, no. 17, pp. 1664–1673, 2019
23. M. Keçeli, S. N. Elliott, Y.-P. Li, M. S. Johnson, C. Cavallotti, Y. Georgievskii, W. H. Green, M. Pelucchi, J. M. Wozniak, A. W. Jasper, *et al.*, "Automated computational thermochemistry for butane oxidation: A prelude to predictive automated combustion kinetics," *Proceedings of the Combustion Institute*, vol. 37, no. 1, pp. 363–371, 2019
24. M. Keçeli, F. Corsetti, C. Campos, J. E. Roman, H. Zhang, Á. Vázquez-Mayagoitia, P. Zapol, and A. F. Wagner, "Siesta-sips: Massively parallel spectrum-slicing eigensolver for an ab initio molecular dynamics package," 2018
25. S. Sarikurt, D. Çakır, M. Keçeli, and C. Sevik, "The influence of surface functionalization on thermal transport and thermoelectric properties of mxene monolayers," *Nanoscale*, vol. 10, no. 18, pp. 8859–8868, 2018
26. Z. Fang, J. Both, S. Li, S. Yue, E. Apra, M. Keçeli, A. F. Wagner, and D. A. Dixon, "Benchmark calculations of energetic properties of groups 4 and 6 transition metal oxide nanoclusters including comparison to density functional theory," *Journal of Chemical Theory and Computation*, vol. 12, no. 8, pp. 3689–3710, 2016
27. M. Keçeli, H. Zhang, P. Zapol, D. A. Dixon, and A. F. Wagner, "Shift-and-invert parallel spectral transformation eigensolver: Massively parallel performance for density-functional based tight-binding," *Journal of Computational Chemistry*, vol. 37, no. 4, pp. 448–459, 2016

28. S. Hirata, M. Keçeli, Y.-y. Ohnishi, O. Sode, and K. Yagi, "Extensivity of energy and electronic and vibrational structure methods for crystals," *Annual Review of Physical Chemistry*, vol. 63, pp. 131–153, 2012
29. O. Sode, M. Keçeli, K. Yagi, and S. Hirata, "Fermi resonance in solid CO₂ under pressure," *The Journal of Chemical Physics*, vol. 138, no. 7, p. 074501, 2013
30. K. Yagi, M. Keçeli, and S. Hirata, "Optimized coordinates for anharmonic vibrational structure theories," *The Journal of Chemical Physics*, vol. 137, no. 20, p. 204118, 2012
31. M. R. Hermes, M. Keçeli, and S. Hirata, "Size-extensive vibrational self-consistent field methods with anharmonic geometry corrections," *The Journal of Chemical Physics*, vol. 136, no. 23, p. 234109, 2012
32. M. Keçeli and S. Hirata, "Size-extensive vibrational self-consistent field method," *The Journal of Chemical Physics*, vol. 135, no. 13, p. 134108, 2011
33. M. Keçeli and S. Hirata, "Fast coupled-cluster singles and doubles for extended systems: Application to the anharmonic vibrational frequencies of polyethylene in the γ approximation," *Physical Review B*, vol. 82, no. 11, p. 115107, 2010
34. S. Hirata, M. Keçeli, and K. Yagi, "First-principles theories for anharmonic lattice vibrations," *The Journal of Chemical Physics*, vol. 133, no. 3, p. 034109, 2010
35. M. Keçeli, S. Hirata, and K. Yagi, "First-principles calculations on anharmonic vibrational frequencies of polyethylene and polyacetylene in the γ approximation," *The Journal of Chemical Physics*, vol. 133, no. 3, p. 034110, 2010
36. M. Keçeli, T. Shiozaki, K. Yagi, and S. Hirata, "Anharmonic vibrational frequencies and vibrationally-averaged structures of key species in hydrocarbon combustion: HCO⁺, HCO, HNO, HOO, HOO⁻, CH₃⁺, and CH₃," *Molecular Physics*, vol. 107, no. 8-12, pp. 1283–1301, 2009
37. O. Sode, M. Keçeli, S. Hirata, and K. Yagi, "Coupled-cluster and many-body perturbation study of energies, structures, and phonon dispersions of solid hydrogen fluoride," *International Journal of Quantum Chemistry*, vol. 109, no. 9, pp. 1928–1939, 2009
38. M. Keçeli, F. Ilday, and M. Oktel, "Ansatz from nonlinear optics applied to trapped Bose-Einstein condensates," *Physical Review A*, vol. 75, no. 3, p. 035601, 2007
39. M. Keçeli and M. Oktel, "Tkachenko modes and structural phase transitions of the vortex lattice of a two-component Bose-Einstein condensate," *Physical Review A*, vol. 73, no. 2, p. 023611, 2006

Book Chapters

- S. Hirata, K. Gilliard, X. He, M. Keçeli, J. Li, M.A. Salim, O. Sode, and K. Yagi, 'Ab initio ice, dry ice, and liquid water,' a chapter in 'Fragmentation: Toward Accurate Calculations on Complex Molecular Systems' edited by M. S. Gordon. (Wiley, Chichester, 2017)
- S. Hirata, O. Sode, M. Keçeli, and T. Shiozaki, 'Electron correlation in solids: delocalized and localized orbital approaches,' a chapter in *Accurate Condensed-Phase Quantum Chemistry* edited by F. Manby, p.129-161. (CRC Press, Boca Raton, 2010)
- S. Hirata, P.-D. Fan, M. Head-Gordon, M. Kamiya, M. Keçeli, T. J. Lee, T. Shiozaki, J. Szczepanski, K. Yagi, M. Vala, and E. F. Valeev, 'Computational interstellar chemistry,' a chapter in *Recent Advances in Spectroscopy: Astrophysical, Theoretical and Experimental Perspective* edited by R. K. Chaudhuri, R. K. Mekkaden, M. V. Raveendran, A. S. Narayanan, p.21-30. (Springer-Verlag, Berlin, 2010)

Other Publications

- M. Keçeli, William H. Green, "Lumping analysis of sulfur pseudospecies in HiFi," Technical report for BP. (2013)
- M. Keçeli and M. Ö. Oktel, "Tkachenko Modes of the Square Vortex Lattice in a two-component Bose-Einstein Condensate," Turk. J. Phys. 30 (4), 245-252. (2006)

Talks, Webinars, and Workshops

- Artificial Intelligence for Molecules and Materials in the Exascale Era, CELS Student Lecture Series, July 16, 2024.
- Hands-on Getting Started on Polaris Workshop, University of Illinois Chicago ACER - ALCF Lighthouse Initiative, June 28, 2024.
- Invited talk, Artificial Intelligence for Computational Chemistry in the Exascale Era, University of Illinois, Chicago, USA, May 28, 2024.
- Technical overview (virtual) of ALCF Resources, University of Illinois Chicago ACER - ALCF Lighthouse Initiative, May 3, 2024.
- Webinar, ALCF Developer Sessions, Interactive High-Performance Computing with ALCF JupyterHub, 2023.
- Invited talk, Machine Learning for Molecules and Materials in the Exascale Era, Stevens Institute of Technology, New Jersey, USA, 2023.
- Invited talk, Machine Learning Potentials for 2D Materials, Molecular Materials Group Meeting, Lemont, Illinois, USA, 2022.
- Invited talk (virtual), Machine Learning and High-Performance Computing for Material Science, Magnetic Properties from First Principles Workshop, Turkey, 2021.
- Invited talk, CNM Theory and Modeling Seminar, Lemont, Illinois, USA, 2021.
- Invited talk (virtual), AI and HPC Journal Club at Argonne: "Distributed Deep Learning and Neural Network Potentials for Quantum Chemistry", 2020.
- Webinar, ALCF Developer Sessions, "Towards Interactive High-Performance Computing with ALCF JupyterHub", 2020.
- Invited, ECP Industrial Council Deep Dive Workshop (virtual) talk: Computational Chemistry for the Exascale Era, 2020.
- Invited colloquium at University of Maryland at Baltimore County Physics Department: Computational Physics in the Exascale Era: Opportunities and Challenges, 2019.
- Contributed talk, ACS national meeting in Orlando, Florida, USA in 2019.
- Invited talk, ELSI/MOLSSI Workshop in Richmond, Virginia, USA in 2018.
- Contributed talk, APS March meeting in Los Angeles, California, USA in 2018.
- Contributed talk, APS March meeting in New Orleans, Louisiana, USA in 2017.
- ALCF Workshop, The Intel Xeon Phi Users Group Meeting, Lemont, Illinois, USA in 2016.
- ALCF Workshop, Scaling Your Science on Mira, Lemont, Illinois, USA in 2016.

- Invited talk at Q-Chem Inc., Pleasanton, California, USA in 2016.
- Contributed talk, APS March meeting in San Antonio, Texas, USA in 2015.
- Contributed talk, ACS national meeting in Denver, Colorado, USA in 2015.
- Poster presentation, PETSc Conference, Lemont, Illinois, USA in 2015.
- Workshop, Argonne Training Program on Extreme Scale Computing, Argonne, Illinois, USA in 2014
- Poster presentation, Midwest Theoretical Chemistry Conference at Evanston, IL, USA in 2014.
- Invited talk at Argonne National Laboratory, Lemont, IL, USA in 2013.
- Poster presentation, MIT Energy Night at Cambridge, MA, USA in 2013.
- Invited talk at Schrödinger Inc, New York, NY, USA in 2013.
- Poster presentation, BP-MIT Conversion Research Program at Cambridge, MA, USA in 2013.
- Invited talk at UC Davis, Davis, California, USA in 2012.
- Invited talk at Stanford University, Stanford, California, USA in 2012.
- Invited talk at UC Irvine, Irvine, California, USA in 2012.
- Invited talk at MIT, Cambridge, Massachusetts, USA in 2012.
- Poster presentation, Quantum Molecular Dynamics Conference, Berkeley, California, USA in 2012.
- Workshop, Building Communities for Software Infrastructure for Sustained Innovation, Arlington, Virginia, USA in 2011.
- Workshop, Petascale Programming Environments and Tools, Urbana, Illinois, USA in 2011.
- Poster presentation, ACS National Meeting, Anaheim, California, USA in 2011.
- Poster presentations, Sanibel Symposium in St. Simons island, Georgia, USA in 2008, 2009, and 2010.
- Contributed talk, APS March meeting in Portland, Oregon, USA in 2010.
- Poster presentation, Conference on Current Trends in Computational Chemistry in Jackson, Mississippi, USA in 2009.
- Poster presentation, The World Congress of WATOC in Sydney, Australia in 2008.
- Contributed talk, İstanbul Statistical Physics Days, İstanbul, Turkey in 2006.
- Summer school, Quantum Phase Transitions and Non-Equilibrium Phenomena in Cold Atomic Gases, Trieste, Italy in 2005.
- Summer school, Quantum Computation at the Atomic Scale, İstanbul, Turkey in 2003.

Software Products

- ChemGraph (Project lead) is an agentic framework for computational chemistry and material science. Available on GitHub.
- Interactive Quantum Chemistry (IQC, lead developer) automates high-throughput workflows and provides an interactive dashboard for computational chemistry and material science. Available on GitHub.

- Quantum Thermochemistry Calculator (QTC, lead developer) automates workflows for accurate thermochemistry calculations. Available on GitHub.
- VALENCE (co-developer) is a quantum chemistry code that provides a massively parallel implementation of variational subspace valence bond method. Available on GitHub.
- Auto-Mech (co-developer) is a software suite that automates high-accuracy quantum chemistry, thermochemistry, and kinetics calculations. Available on GitHub.
- NWChemEx (co-developer) is an electronic structure package focusing on high-performance computing and reduced scaling methodologies. Available on GitHub. Some of the repos will be released within this year.
- ezHPC (developer) is a set of Jupyter notebooks that provides interactive features for ALCF supercomputers. Available on GitHub.
- miniGAP (co-developer) is a proxy application for molecular and materials property prediction using the Gaussian Process Approximation. Available on GitHub.
- NinjaParser (developer) is a Python script to parse ninja log files to identify codes with longest compilation times. Available on GitHub.
- Contributed to various other open-source projects, such as TiledArray, MADNESS, Google/FFN, and Balsam.

Research Highlights

High-performance computing and artificial intelligence at ANL

Jan 2014 - Present

- Application and evaluation of large language models (LLMs) for scientific tasks.
 - Developed an agentic framework for computational chemistry simulations.
 - Created an LLM evaluation benchmark for computational chemistry and initiated a HuggingFace leaderboard.
 - Implemented a Retrieval-Augmented Generation (RAG) pipeline for ALCF documentation and evaluated different large-language models.
 - Generated (both manually and automated) multiple-choice questions for chemistry, physics, and material science domains for LLMs.
 - Evaluated different retrieval augmentation and fine-tuning methods to customize LLMs for domain-specific tasks, enhancing their accuracy and relevance in generating contextually appropriate responses.
 - Explored utilization of LLM agents for enhancing scientific workflows, focusing on their ability to automate data analysis, streamline complex computational tasks, and improve the overall efficiency.
- Argonne point-of-contact and a co-developer for the NWChemEx Project, an ECP funded application software for quantum chemistry.
 - Contributed to the implementation of the C++ infrastructure code responsible for caching and serialization mechanisms.
 - Contributed to the design and implementation of the user-interface based on Python bindings.
 - Actively participating in the development and maintenance activities such as adding features, fixing issues, reviewing pull requests, and improving documentation.
- Developing machine learning potentials for material science and chemistry.

- Developed Gaussian Approximation Potentials (GAPs) for two-dimensional materials and verified their accuracy for thermal properties. Published the models and training data on GitHub.
- Compared the accuracy and training efficiency of GAPs with neural network based potentials.
- Implemented distributed training for equivariant graph neural networks, benchmarked and optimized the performance on Polaris.
- Developed neural network potentials for salt mixtures using density functional theory calculations.
- Co-developer of VALENCE quantum chemistry code based on variational subspace valence bond method.
 - Developed Vtools that can automate input preparation and visualization.
 - Contributed to optimization and benchmarking of the code on Intel-KNL nodes of Theta supercomputer.
- Reduced training time of image segmentation code for connectomics.
 - Enabled and optimized data parallel training of deep convolutional neural networks for 3D image segmentation by using Horovod and hyperparameter optimization.
 - Reduced the training time from seven days to less than 5 hours by using 1024 Intel KNL CPUs on ALCF Theta supercomputer.
 - Parallelized the preprocessing step with mpi4py to reduce the computational time from 4 hours to five minutes.
- Co-developer of Auto-Mech software suite for computational thermochemistry and kinetics.
 - Developed Quantum Thermochemistry Calculator to automate complex workflows for thermochemistry calculations.
 - Working on the curation and optimization of a database to efficiently store and manage the results generated by Auto-Mech.
 - Working on porting Auto-Mech codes to Polaris and Aurora and optimizing MESS code with multiprecision algorithms.
- Development, testing and benchmarking of sparse and dense eigensolvers for quantum chemistry applications.
 - Developed shift-and-invert spectrum slicing eigensolver (SIPs) based on PETSc and SLEPc libraries.
 - Demonstrated strong scaling beyond 200,000 cores and enabled density functional tight-binding (DFTB) simulations of more than 100,000 atoms. This work was published on the cover of Journal of Computational Chemistry and highlighted on ALCF newsbytes.
 - Integrated the massively parallel sparse eigensolver, SIPs with SIESTA ab initio molecular dynamics package and demonstrated higher strong scaling efficiency and reduced time to solution compared to default solver, ScaLAPACK. The code has been ported and tested on BG/Q nodes at ALCF-Mira, Intel-Haswell nodes at NERSC-Cori, and Intel-KNL nodes at ALCF-Theta supercomputers.
 - Developed a Python code, PSCF, based on mpi4py, petsc4py and slepc4py for semi-empirical molecular orbital calculations.
- Performed benchmark coupled-cluster and DFT calculations with NWChem and Molpro for accurate thermochemistry of transition metal oxide clusters. Some of these calculations involve more than 1300 basis functions and were run on supercomputers.

Chemical engineering and software development at MIT

July 2012 - Jan 2014

- Developed a “fit-for-purpose” model using kinetic model reduction and lumping analysis in a proprietary hydrotreating reactor model as part of BP-MIT Advanced Conversion Research Program. The reductions in the model can eliminate various feed characterization experiments and reduce capital costs significantly.

- Served as a member of the software development team of reaction mechanism generator (RMG) program, which constructs kinetic models composed of elementary chemical reaction steps. Contributed to the coding, testing and bug-fixing efforts before the major release of the Java version and lead the parallelization of the Python version. See <http://github.com/ReactionMechanismGenerator/RMG-Java>.

Quantum chemistry at UIUC and UF

2007 - 2012

- Developed a general scheme to calculate anharmonic vibrational frequencies and vibrationally-averaged structures of molecules at the complete-correlation complete-basis-set limit using a novel hybrid potential energy surface representation that combines the merits of grid and Taylor expansion based representations.
- Utilized coupled-cluster singles and doubles with a noniterative triples [CCSD(T)] correction for solid hydrogen fluoride by employing a type of fragment molecular orbital (FMO) method based on a truncated many-body expansion of the energies of molecular crystals.
- Introduced the mod- n approximation (uniform downsampling of the wave vectors in Brillouin-zone integrations) for crystal-orbital coupled-cluster singles and doubles (CCSD) method that reduces the computational cost more than 500 times.
- Invented the diagonalization-free size-extensive vibrational self-consistent field (XVSCF) method for molecules, which is the fastest variational approach to include anharmonicity in vibrational calculations.
- Wrote a standalone computer program (Fortran 95) that can generate a quartic force field potential energy surface and solve the vibrational Schrödinger equation based on Watson Hamiltonian. Implemented methods are: VSCF, virtual VSCF, VPT2, VMP2, HO-VCI, VSCF-VCI, XVSCF, XVPT2, and XVMP2.
 - M. Keçeli, MaVi, University of Illinois at Urbana-Champaign, (2011).
- Applications of high-level quantum chemical methods for key species in hydrocarbon combustion, such as HCO^+ , HCO , HNO , HOO , HOO , CH_3^+ , and CH_3 , and extended systems such as polyethylene, polyacetylene, solid hydrogen fluoride, and solid carbon dioxide.
- Contributed to the development of a robust and general optimization algorithm for vibrational coordinates to reduce the cost of VCI calculations.

Computational atomic physics at Bilkent

2004 - 2006

- Demonstrated that a simple analytical ansatz, which has been used to describe the intensity profile of the similariton laser, can be utilized as a variational wave function to solve the Gross-Pitaevskii equation for a wide range of interaction parameters.
- Derived dispersion relations for collective modes of a vortex lattice formed in a Bose-Einstein condensate, numerically calculated elastic constants for different lattice geometries and identified structural phase transitions.

Outreach and Teaching

- Served as a mentor for the Student Inquiry and Research program of Illinois Mathematics and Science Academy in 2024-2025.
- Participated in ALCF outreach events such as Introduce a Girl to Engineering Day event in 2024 and 2025.
- Served as a mentor and a project leader for Introduction to HPC bootcamp in 2023 and in 2025.
- Mentored three students for the Sustainable Research Pathways program in the summer of 2023.

- Participated in "Breakthroughs Enabled by Computing" activity at the Argonne Open House in 2023.
- Prepared and presented webinars on interactive computing for HPC using Jupyter environment in 2021 and in 2023.
- Prepared and presented lectures during annual Hour-of-Code activities at local elementary and middle schools annually since 2019.
- Served as a jury for middle school annual science fair projects at the Avery Coonley School since 2018.
- Teaching assistant for the freshman physics courses at University of Florida in 2006 and 2007.
- Teaching assistant for the junior level quantum mechanics at Bilkent University in 2005.
- Teaching assistant for the freshman physics courses at Bilkent University in 2004.
- Tutor for the freshman physics course at Bilkent University in 2003.
- Volunteer tutor of science lectures for underprivileged lower and middle school students in Ankara, 2002-2003.